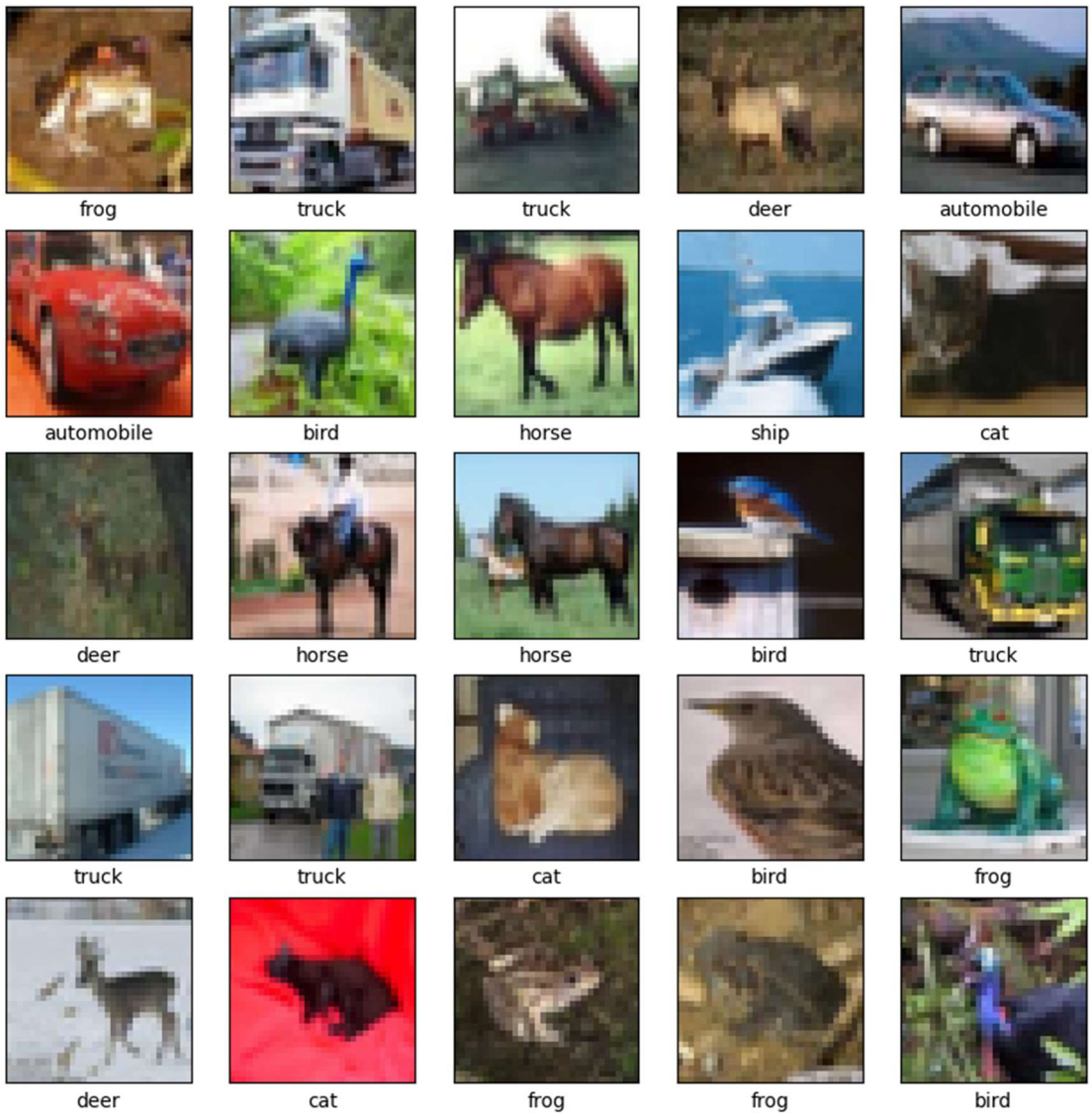
```python
import tensorflow as tf
from tensorflow.keras.datasets import cifar10
from tensorflow.keras.models import Sequential
from tensorflow.keras.layers import Flatten, Dense, Dropout, Conv2D,
MaxPooling2D
from tensorflow.keras.utils import to_categorical
from sklearn.model_selection import train_test_split
from sklearn.metrics import accuracy_score, classification_report,
confusion_matrix
import matplotlib.pyplot as plt
import pandas as pd
import numpy as np
import seaborn as sns

# Step 1: Load and Explore the Data (10 points)
(x_train_full, y_train_full), (x_test, y_test) = cifar10.load_data()

# Determine training and testing data shapes.
print("Training data shape:", x_train_full.shape)  # (50000, 32, 32, 3)
print("Test data shape:", x_test.shape)  # (10000, 32, 32, 3)

# Define class names.
class_names = ['airplane', 'automobile', 'bird', 'cat', 'deer', 'dog',
'frog', 'horse', 'ship', 'truck']

# Visualize 25 samples.
plt.figure(figsize=(10, 10))
for i in range(25):
    plt.subplot(5, 5, i+1)
    plt.xticks([])
    plt.yticks([])
    plt.grid(False)
    plt.imshow(x_train_full[i])
    plt.xlabel(class_names[y_train_full[i][0]])
plt.show()

# Generate class distribution.
unique, counts = np.unique(y_train_full, return_counts=True)

# Create a DataFrame.
df_distribution = pd.DataFrame({
    'Class': class_names,
    'Count': counts
})

# Display the DataFrame.
print("Class Distribution:")
print(df_distribution)
```

```
Training data shape: (50000, 32, 32, 3)
Test data shape: (10000, 32, 32, 3)
```




```
Class Distribution:
        Class  Count
0    airplane   5000
1  automobile   5000
2        bird   5000
3         cat   5000
4        deer   5000
5         dog   5000
6        frog   5000
7       horse   5000
8        ship   5000
9       truck   5000
```

```python
# Step 2: Split the Data for Training and Testing (10 points)
# Create validation set (20% of training data)
x_train, x_val, y_train, y_val = train_test_split(x_train_full, y_train_full,
test_size=0.2, random_state=42)

# Normalize data (scale pixels to [0, 1]).
x_train = x_train.astype('float32') / 255.0
x_val = x_val.astype('float32') / 255.0
x_test = x_test.astype('float32') / 255.0

# One-hot encode labels.
y_train = to_categorical(y_train, 10)
y_val = to_categorical(y_val, 10)
y_test = to_categorical(y_test, 10)

# Step 3: Train the Model Using Simple Artificial Neural Network (ANN)
# Select and Build the Model (15 points)
ann_model = Sequential([
    Flatten(input_shape=(32, 32, 3)),  # Flatten to 1D (32*32*3 = 3072)
    Dense(512, activation='relu'),
    Dropout(0.2),
    Dense(256, activation='relu'),
    Dropout(0.2),
    Dense(10, activation='softmax')
])
ann_model.summary()

# Train the Model (5 points)
ann_model.compile(optimizer='adam', loss='categorical_crossentropy',
metrics=['accuracy'])
history_ann = ann_model.fit(x_train, y_train, epochs=30, batch_size=128,
validation_data=(x_val, y_val))

# Plot ANN training history.
plt.plot(history_ann.history['accuracy'], label='ANN Train Accuracy')
plt.plot(history_ann.history['val_accuracy'], label='ANN Val Accuracy')
plt.xlabel('Epoch')
plt.ylabel('Accuracy')
plt.legend()
plt.title('ANN Training History')
plt.show()

# Make Predictions (5 points)
ann_predictions = ann_model.predict(x_test)
ann_pred_classes = np.argmax(ann_predictions, axis=1)
y_test_classes = np.argmax(y_test, axis=1)
```

```python
# Evaluate the Result (5 points)
ann_accuracy = accuracy_score(y_test_classes, ann_pred_classes)
print("ANN Test Accuracy:", ann_accuracy)
print(classification_report(y_test_classes, ann_pred_classes,
target_names=class_names))

# Generate ANN confusion matrix.
cm_ann = confusion_matrix(y_test_classes, ann_pred_classes)
sns.heatmap(cm_ann, annot=True, fmt='d', cmap='Blues',
xticklabels=class_names, yticklabels=class_names)
plt.title('ANN Confusion Matrix')
plt.show()
```

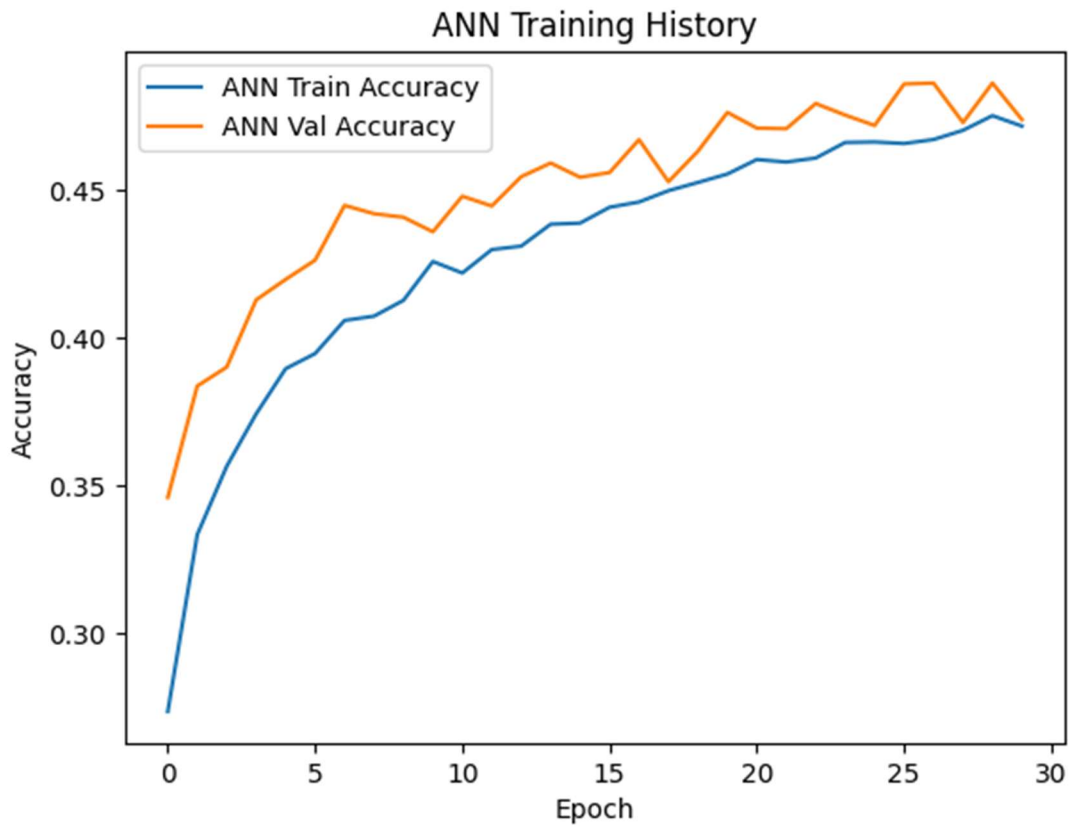
```
Model: "sequential_12"
```

Layer (type)	Output Shape	Param #
flatten_12 (Flatten)	(None, 3072)	0
dense_30 (Dense)	(None, 512)	1,573,376
dropout_18 (Dropout)	(None, 512)	0
dense_31 (Dense)	(None, 256)	131,328
dropout_19 (Dropout)	(None, 256)	0
dense_32 (Dense)	(None, 10)	2,570

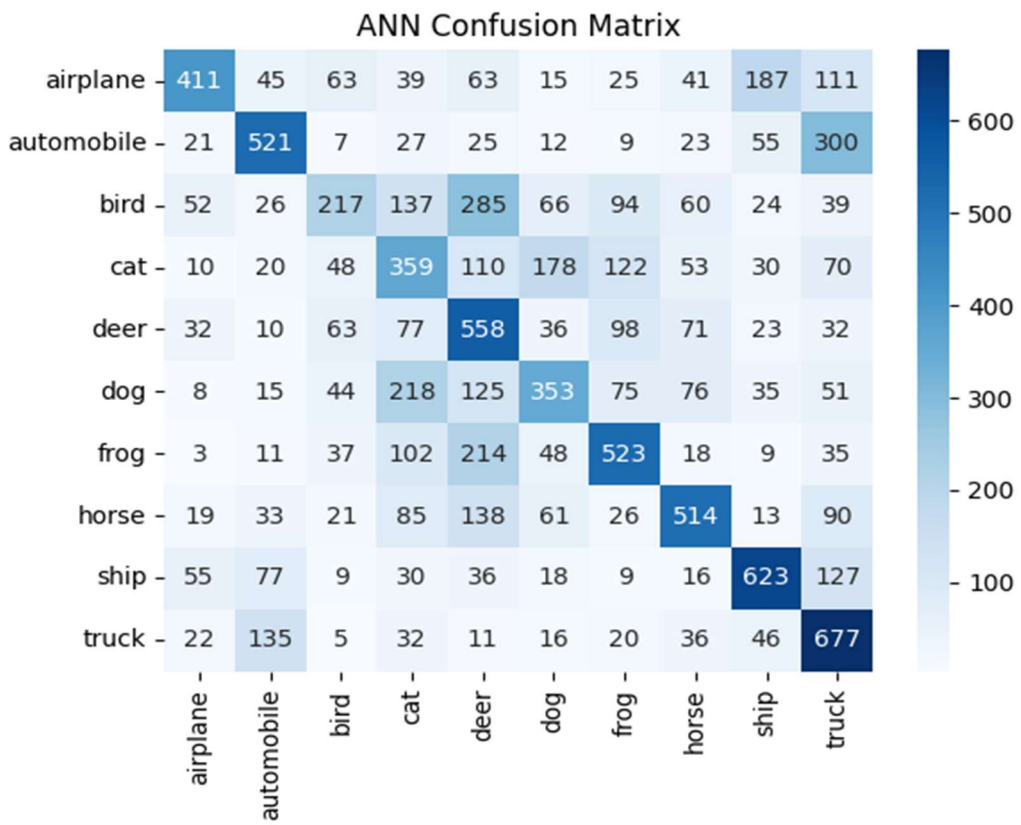
```
Total params: 1,707,274 (6.51 MB)

Trainable params: 1,707,274 (6.51 MB)

Non-trainable params: 0 (0.00 B)
```

```
Epoch 1/30
313/313 ━━━━━━━━━━━━━━━━━━━━ 7s 20ms/step - accuracy: 0.2246 - loss: 2.1844 -
val_accuracy: 0.3461 - val_loss: 1.8183
Epoch 2/30
313/313 ━━━━━━━━━━━━━━━━━━━━ 6s 19ms/step - accuracy: 0.3237 - loss: 1.8610 -
val_accuracy: 0.3837 - val_loss: 1.7477
Epoch 3/30
313/313 ━━━━━━━━━━━━━━━━━━━━ 6s 20ms/step - accuracy: 0.3513 - loss: 1.7893 -
val_accuracy: 0.3901 - val_loss: 1.7147
Epoch 4/30
313/313 ━━━━━━━━━━━━━━━━━━━━ 6s 19ms/step - accuracy: 0.3698 - loss: 1.7541 -
val_accuracy: 0.4128 - val_loss: 1.6440
Epoch 5/30
313/313 ━━━━━━━━━━━━━━━━━━━━ 6s 20ms/step - accuracy: 0.3877 - loss: 1.7049 -
val_accuracy: 0.4197 - val_loss: 1.6207
Epoch 6/30
313/313 ━━━━━━━━━━━━━━━━━━━━ 6s 19ms/step - accuracy: 0.3940 - loss: 1.6797 -
val_accuracy: 0.4262 - val_loss: 1.6023
Epoch 7/30
313/313 ━━━━━━━━━━━━━━━━━━━━ 6s 20ms/step - accuracy: 0.4086 - loss: 1.6434 -
val_accuracy: 0.4447 - val_loss: 1.5773
Epoch 8/30
313/313 ━━━━━━━━━━━━━━━━━━━━ 6s 20ms/step - accuracy: 0.4078 - loss: 1.6414 -
val_accuracy: 0.4419 - val_loss: 1.5599
Epoch 9/30
313/313 ━━━━━━━━━━━━━━━━━━━━ 6s 19ms/step - accuracy: 0.4137 - loss: 1.6173 -
val_accuracy: 0.4407 - val_loss: 1.5801
Epoch 10/30
313/313 ━━━━━━━━━━━━━━━━━━━━ 6s 19ms/step - accuracy: 0.4244 - loss: 1.6046 -
val_accuracy: 0.4358 - val_loss: 1.5669
Epoch 11/30
313/313 ━━━━━━━━━━━━━━━━━━━━ 6s 19ms/step - accuracy: 0.4236 - loss: 1.5879 -
val_accuracy: 0.4478 - val_loss: 1.5410
Epoch 12/30
313/313 ━━━━━━━━━━━━━━━━━━━━ 6s 19ms/step - accuracy: 0.4282 - loss: 1.5800 -
val_accuracy: 0.4445 - val_loss: 1.5567
Epoch 13/30
313/313 ━━━━━━━━━━━━━━━━━━━━ 6s 19ms/step - accuracy: 0.4312 - loss: 1.5767 -
val_accuracy: 0.4544 - val_loss: 1.5250
Epoch 14/30
313/313 ━━━━━━━━━━━━━━━━━━━━ 6s 19ms/step - accuracy: 0.4424 - loss: 1.5650 -
val_accuracy: 0.4590 - val_loss: 1.5198
Epoch 15/30
313/313 ━━━━━━━━━━━━━━━━━━━━ 6s 19ms/step - accuracy: 0.4428 - loss: 1.5455 -
val_accuracy: 0.4542 - val_loss: 1.5284
Epoch 16/30
313/313 ━━━━━━━━━━━━━━━━━━━━ 6s 19ms/step - accuracy: 0.4457 - loss: 1.5488 -
val_accuracy: 0.4558 - val_loss: 1.5305
```

```
Epoch 17/30
313/313 ━━━━━━━━━━━━━━━━━━━━ 6s 19ms/step - accuracy: 0.4431 - loss: 1.5389 -
val_accuracy: 0.4669 - val_loss: 1.4961
Epoch 18/30
313/313 ━━━━━━━━━━━━━━━━━━━━ 6s 19ms/step - accuracy: 0.4530 - loss: 1.5241 -
val_accuracy: 0.4527 - val_loss: 1.5343
Epoch 19/30
313/313 ━━━━━━━━━━━━━━━━━━━━ 6s 19ms/step - accuracy: 0.4533 - loss: 1.5194 -
val_accuracy: 0.4630 - val_loss: 1.5075
Epoch 20/30
313/313 ━━━━━━━━━━━━━━━━━━━━ 6s 19ms/step - accuracy: 0.4578 - loss: 1.5109 -
val_accuracy: 0.4761 - val_loss: 1.4843
Epoch 21/30
313/313 ━━━━━━━━━━━━━━━━━━━━ 6s 19ms/step - accuracy: 0.4606 - loss: 1.5032 -
val_accuracy: 0.4708 - val_loss: 1.4827
Epoch 22/30
313/313 ━━━━━━━━━━━━━━━━━━━━ 6s 19ms/step - accuracy: 0.4677 - loss: 1.4824 -
val_accuracy: 0.4706 - val_loss: 1.4787
Epoch 23/30
313/313 ━━━━━━━━━━━━━━━━━━━━ 6s 19ms/step - accuracy: 0.4635 - loss: 1.4937 -
val_accuracy: 0.4791 - val_loss: 1.4910
Epoch 24/30
313/313 ━━━━━━━━━━━━━━━━━━━━ 6s 19ms/step - accuracy: 0.4694 - loss: 1.4805 -
val_accuracy: 0.4752 - val_loss: 1.4853
Epoch 25/30
313/313 ━━━━━━━━━━━━━━━━━━━━ 6s 19ms/step - accuracy: 0.4667 - loss: 1.4851 -
val_accuracy: 0.4717 - val_loss: 1.4997
Epoch 26/30
313/313 ━━━━━━━━━━━━━━━━━━━━ 6s 19ms/step - accuracy: 0.4702 - loss: 1.4803 -
val_accuracy: 0.4857 - val_loss: 1.4628
Epoch 27/30
313/313 ━━━━━━━━━━━━━━━━━━━━ 6s 19ms/step - accuracy: 0.4610 - loss: 1.4786 -
val_accuracy: 0.4860 - val_loss: 1.4525
Epoch 28/30
313/313 ━━━━━━━━━━━━━━━━━━━━ 6s 19ms/step - accuracy: 0.4665 - loss: 1.4749 -
val_accuracy: 0.4727 - val_loss: 1.4875
Epoch 29/30
313/313 ━━━━━━━━━━━━━━━━━━━━ 6s 19ms/step - accuracy: 0.4763 - loss: 1.4660 -
val_accuracy: 0.4860 - val_loss: 1.4523
Epoch 30/30
313/313 ━━━━━━━━━━━━━━━━━━━━ 6s 19ms/step - accuracy: 0.4756 - loss: 1.4586 -
val_accuracy: 0.4737 - val_loss: 1.4774
```




```
313/313 ━━━━━━━━━━━━━━━━━━━━ 1s 2ms/step
ANN Test Accuracy: 0.4756
              precision    recall  f1-score   support

    airplane       0.65      0.41      0.50      1000
  automobile       0.58      0.52      0.55      1000
        bird       0.42      0.22      0.29      1000
         cat       0.32      0.36      0.34      1000
        deer       0.36      0.56      0.44      1000
         dog       0.44      0.35      0.39      1000
        frog       0.52      0.52      0.52      1000
       horse       0.57      0.51      0.54      1000
        ship       0.60      0.62      0.61      1000
       truck       0.44      0.68      0.53      1000

    accuracy                           0.48     10000
   macro avg       0.49      0.48      0.47     10000
weighted avg       0.49      0.48      0.47     10000
```



In the training history chart, ANN Training and Validation accuracy began to converge after 30 epochs. ANN Testing accuracy garnered a f1-score of 0.48 approximately. In the classification matrix, the 'bird' image recognition seemed the least reliable (f1-score: 0.29), and the 'ship' image recognition seemed the most reliable (f1-score: 0.61). In the confusion matrix, the 'bird' image was interpreted more as a 'deer', which is surprising. Some potentially understandable confusion between 'cat' and 'dog' images, as well as 'automobile' and 'truck' images, were noted.

```python
# Step 4: Train a CNN and Check the Performance
# Select and Build the Model (20 points)
cnn_model = Sequential([
    Conv2D(32, (3, 3), activation='relu', input_shape=(32, 32, 3)),
    MaxPooling2D((2, 2)),
    Conv2D(64, (3, 3), activation='relu'),
    MaxPooling2D((2, 2)),
    Conv2D(64, (3, 3), activation='relu'),
    Flatten(),
    Dense(64, activation='relu'),
    Dropout(0.5),
    Dense(10, activation='softmax')
])
cnn_model.summary()

# Train the Model (5 points)
cnn_model.compile(optimizer='adam', loss='categorical_crossentropy',
metrics=['accuracy'])
history_cnn = cnn_model.fit(x_train, y_train, epochs=30, batch_size=128,
validation_data=(x_val, y_val))

# Plot CNN training history.
plt.plot(history_cnn.history['accuracy'], label='CNN Train Accuracy')
plt.plot(history_cnn.history['val_accuracy'], label='CNN Val Accuracy')
plt.xlabel('Epoch')
plt.ylabel('Accuracy')
plt.legend()
plt.title('CNN Training History')
plt.show()

# Make Predictions (10 points)
cnn_predictions = cnn_model.predict(x_test)
cnn_pred_classes = np.argmax(cnn_predictions, axis=1)

# Evaluate the Result (10 points)
cnn_accuracy = accuracy_score(y_test_classes, cnn_pred_classes)
print("CNN Test Accuracy:", cnn_accuracy)
print(classification_report(y_test_classes, cnn_pred_classes,
target_names=class_names))

# Generate CNN confusion matrix.
cm_cnn = confusion_matrix(y_test_classes, cnn_pred_classes)
sns.heatmap(cm_cnn, annot=True, fmt='d', cmap='Blues',
xticklabels=class_names, yticklabels=class_names)
plt.title('CNN Confusion Matrix')
plt.show()
```

Model: "sequential_13"

Layer (type)	Output Shape	Param #
conv2d_18 (Conv2D)	(None, 30, 30, 32)	896
max_pooling2d_12 (MaxPooling2D)	(None, 15, 15, 32)	0
conv2d_19 (Conv2D)	(None, 13, 13, 64)	18,496
max_pooling2d_13 (MaxPooling2D)	(None, 6, 6, 64)	0
conv2d_20 (Conv2D)	(None, 4, 4, 64)	36,928
flatten_13 (Flatten)	(None, 1024)	0
dense_33 (Dense)	(None, 64)	65,600
dropout_20 (Dropout)	(None, 64)	0
dense_34 (Dense)	(None, 10)	650

Total params: 122,570 (478.79 KB)

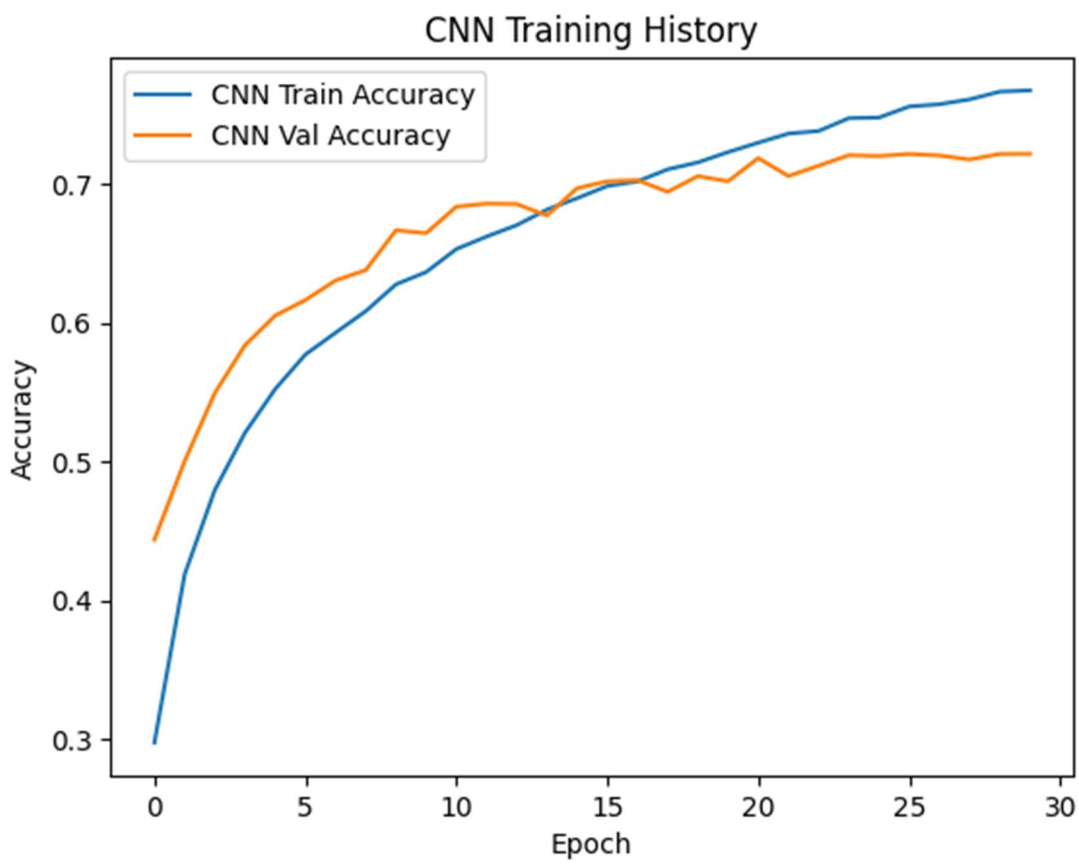
Trainable params: 122,570 (478.79 KB)

Non-trainable params: 0 (0.00 B)

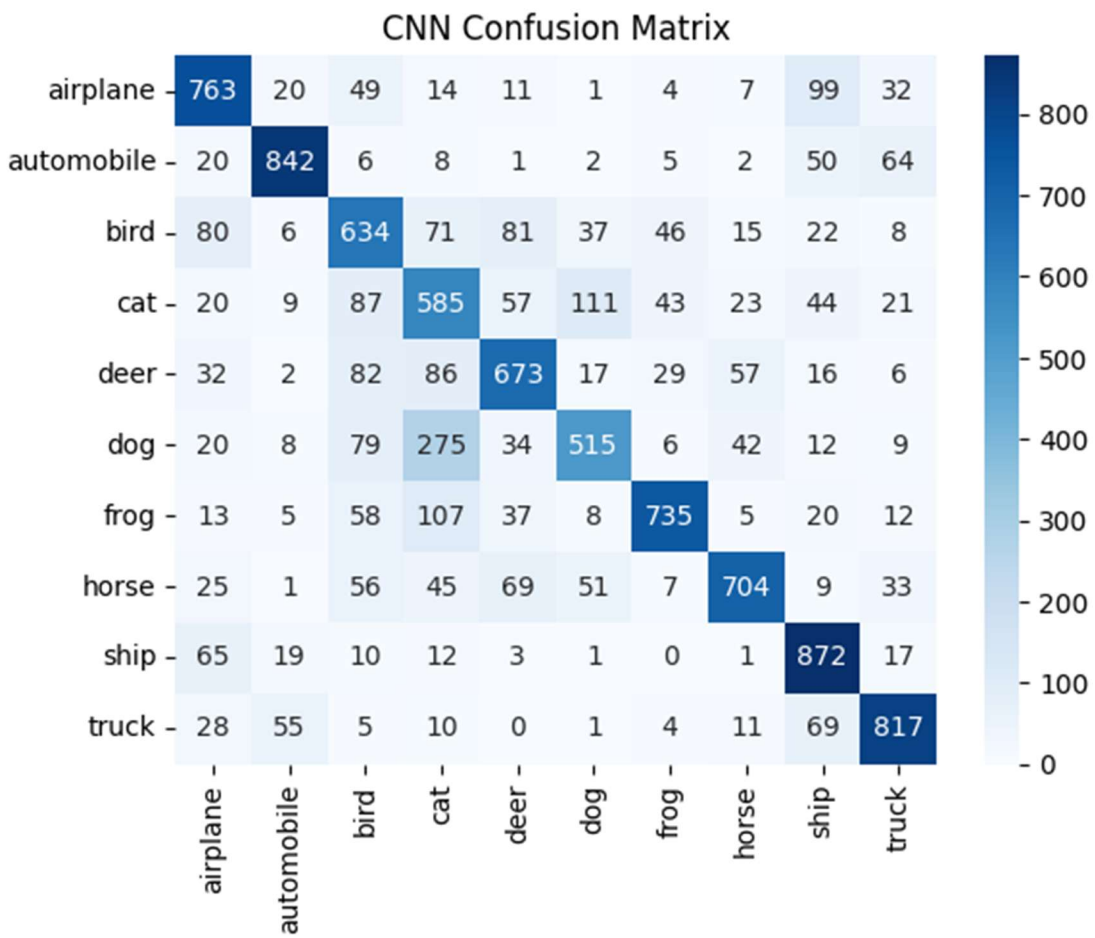
```
Epoch 1/30
313/313 ━━━━━━━━━━━━━━━━━━━━ 12s 33ms/step - accuracy: 0.2283 - loss: 2.0374
- val_accuracy: 0.4439 - val_loss: 1.5317
Epoch 2/30
313/313 ━━━━━━━━━━━━━━━━━━━━ 11s 34ms/step - accuracy: 0.4059 - loss: 1.6145
- val_accuracy: 0.5004 - val_loss: 1.3662
Epoch 3/30
313/313 ━━━━━━━━━━━━━━━━━━━━ 11s 36ms/step - accuracy: 0.4657 - loss: 1.4600
- val_accuracy: 0.5495 - val_loss: 1.2498
Epoch 4/30
313/313 ━━━━━━━━━━━━━━━━━━━━ 12s 37ms/step - accuracy: 0.5126 - loss: 1.3590
- val_accuracy: 0.5840 - val_loss: 1.1611
Epoch 5/30
313/313 ━━━━━━━━━━━━━━━━━━━━ 11s 36ms/step - accuracy: 0.5471 - loss: 1.2720
- val_accuracy: 0.6053 - val_loss: 1.1074
Epoch 6/30
313/313 ━━━━━━━━━━━━━━━━━━━━ 10s 31ms/step - accuracy: 0.5758 - loss: 1.2078
- val_accuracy: 0.6165 - val_loss: 1.0709
Epoch 7/30
313/313 ━━━━━━━━━━━━━━━━━━━━ 11s 36ms/step - accuracy: 0.5946 - loss: 1.1496
- val_accuracy: 0.6306 - val_loss: 1.0351
```

```
Epoch 8/30
313/313 ━━━━━━━━━━━━━━━━━━━━ 12s 39ms/step - accuracy: 0.6090 - loss: 1.1057
- val_accuracy: 0.6381 - val_loss: 1.0027
Epoch 9/30
313/313 ━━━━━━━━━━━━━━━━━━━━ 11s 36ms/step - accuracy: 0.6229 - loss: 1.0670
- val_accuracy: 0.6668 - val_loss: 0.9552
Epoch 10/30
313/313 ━━━━━━━━━━━━━━━━━━━━ 15s 49ms/step - accuracy: 0.6384 - loss: 1.0279
- val_accuracy: 0.6647 - val_loss: 0.9551
Epoch 11/30
313/313 ━━━━━━━━━━━━━━━━━━━━ 16s 52ms/step - accuracy: 0.6585 - loss: 0.9801
- val_accuracy: 0.6838 - val_loss: 0.8996
Epoch 12/30
313/313 ━━━━━━━━━━━━━━━━━━━━ 15s 46ms/step - accuracy: 0.6636 - loss: 0.9595
- val_accuracy: 0.6860 - val_loss: 0.8864
Epoch 13/30
313/313 ━━━━━━━━━━━━━━━━━━━━ 11s 34ms/step - accuracy: 0.6679 - loss: 0.9435
- val_accuracy: 0.6857 - val_loss: 0.8812
Epoch 14/30
313/313 ━━━━━━━━━━━━━━━━━━━━ 12s 37ms/step - accuracy: 0.6804 - loss: 0.9062
- val_accuracy: 0.6774 - val_loss: 0.9313
Epoch 15/30
313/313 ━━━━━━━━━━━━━━━━━━━━ 12s 39ms/step - accuracy: 0.6923 - loss: 0.8843
- val_accuracy: 0.6971 - val_loss: 0.8825
Epoch 16/30
313/313 ━━━━━━━━━━━━━━━━━━━━ 12s 38ms/step - accuracy: 0.6967 - loss: 0.8651
- val_accuracy: 0.7021 - val_loss: 0.8613
Epoch 17/30
313/313 ━━━━━━━━━━━━━━━━━━━━ 11s 37ms/step - accuracy: 0.7038 - loss: 0.8316
- val_accuracy: 0.7029 - val_loss: 0.8580
Epoch 18/30
313/313 ━━━━━━━━━━━━━━━━━━━━ 12s 37ms/step - accuracy: 0.7109 - loss: 0.8232
- val_accuracy: 0.6945 - val_loss: 0.8773
Epoch 19/30
313/313 ━━━━━━━━━━━━━━━━━━━━ 13s 42ms/step - accuracy: 0.7163 - loss: 0.8001
- val_accuracy: 0.7058 - val_loss: 0.8547
Epoch 20/30
313/313 ━━━━━━━━━━━━━━━━━━━━ 12s 38ms/step - accuracy: 0.7252 - loss: 0.7913
- val_accuracy: 0.7021 - val_loss: 0.8672
Epoch 21/30
313/313 ━━━━━━━━━━━━━━━━━━━━ 11s 36ms/step - accuracy: 0.7329 - loss: 0.7599
- val_accuracy: 0.7188 - val_loss: 0.8295
Epoch 22/30
313/313 ━━━━━━━━━━━━━━━━━━━━ 12s 38ms/step - accuracy: 0.7379 - loss: 0.7470
- val_accuracy: 0.7059 - val_loss: 0.8885
Epoch 23/30
313/313 ━━━━━━━━━━━━━━━━━━━━ 13s 40ms/step - accuracy: 0.7412 - loss: 0.7376
- val_accuracy: 0.7133 - val_loss: 0.8401
```

```
Epoch 24/30
313/313 ━━━━━━━━━━━━━━━━━━━━ 12s 39ms/step - accuracy: 0.7504 - loss: 0.7152
- val_accuracy: 0.7209 - val_loss: 0.8367
Epoch 25/30
313/313 ━━━━━━━━━━━━━━━━━━━━ 11s 35ms/step - accuracy: 0.7537 - loss: 0.6994
- val_accuracy: 0.7203 - val_loss: 0.8256
Epoch 26/30
313/313 ━━━━━━━━━━━━━━━━━━━━ 12s 38ms/step - accuracy: 0.7569 - loss: 0.6792
- val_accuracy: 0.7218 - val_loss: 0.8479
Epoch 27/30
313/313 ━━━━━━━━━━━━━━━━━━━━ 12s 38ms/step - accuracy: 0.7594 - loss: 0.6825
- val_accuracy: 0.7207 - val_loss: 0.8416
Epoch 28/30
313/313 ━━━━━━━━━━━━━━━━━━━━ 12s 38ms/step - accuracy: 0.7682 - loss: 0.6475
- val_accuracy: 0.7178 - val_loss: 0.8370
Epoch 29/30
313/313 ━━━━━━━━━━━━━━━━━━━━ 11s 36ms/step - accuracy: 0.7693 - loss: 0.6500
- val_accuracy: 0.7218 - val_loss: 0.8501
Epoch 30/30
313/313 ━━━━━━━━━━━━━━━━━━━━ 12s 38ms/step - accuracy: 0.7693 - loss: 0.6473
- val_accuracy: 0.7219 - val_loss: 0.8612
```



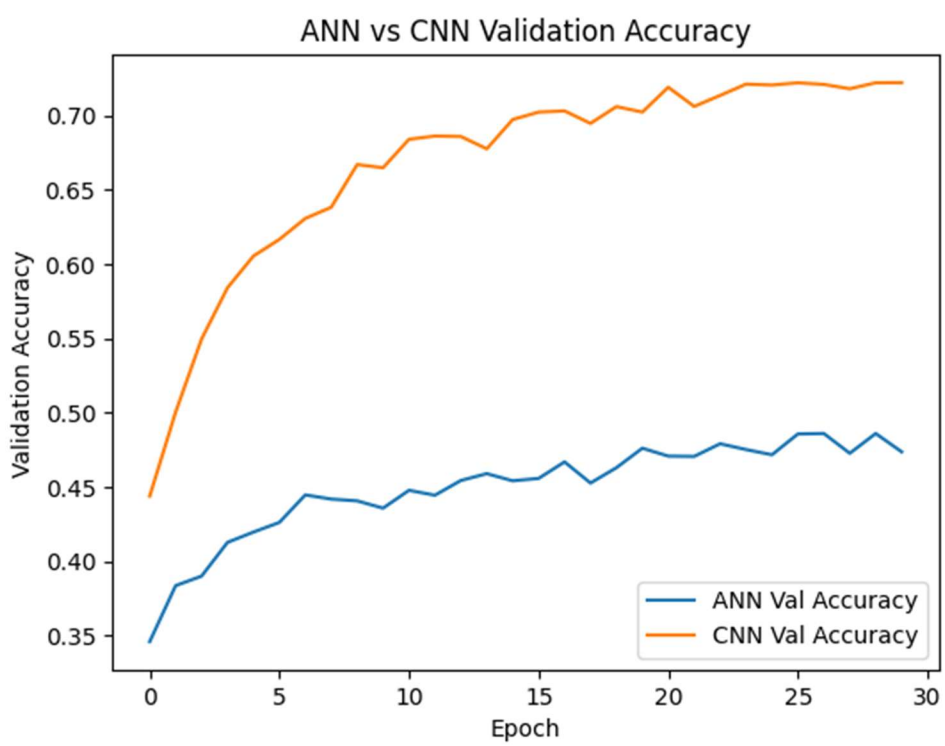
```
313/313 ━━━━━━━━━━━━━━━━━━━━ 2s 5ms/step
CNN Test Accuracy: 0.714
              precision    recall  f1-score   support

    airplane       0.72      0.76      0.74      1000
  automobile       0.87      0.84      0.86      1000
        bird       0.59      0.63      0.61      1000
         cat       0.48      0.58      0.53      1000
        deer       0.70      0.67      0.68      1000
         dog       0.69      0.52      0.59      1000
        frog       0.84      0.73      0.78      1000
       horse       0.81      0.70      0.75      1000
        ship       0.72      0.87      0.79      1000
       truck       0.80      0.82      0.81      1000

    accuracy                           0.71     10000
   macro avg       0.72      0.71      0.71     10000
weighted avg       0.72      0.71      0.71     10000
```



In the training history chart, CNN Training accuracy eventually exceeded the Validation accuracy after 30 epochs, probably due to the larger sample size. CNN Testing accuracy garnered a f1-score of 0.71 approximately. In the classification matrix, the 'cat' image recognition seemed the least reliable (f1-score: 0.53), and the 'automobile' image recognition seemed the most reliable (f1-score: 0.86). In the confusion matrix, the 'cat' image was sometimes confused with a 'dog', which is potentially understandable.

```
# Step 5: Compare the Two Models (5 points)
print(f"ANN Test Accuracy: {ann_accuracy:.4f}")
print(f"CNN Test Accuracy: {cnn_accuracy:.4f}")
imp_accuracy = cnn_accuracy - ann_accuracy
print(f"Improvement with CNN: {imp_accuracy:.4f}")

# Comparatively plot validation accuracies.
plt.plot(history_ann.history['val_accuracy'], label='ANN Val Accuracy')
plt.plot(history_cnn.history['val_accuracy'], label='CNN Val Accuracy')
plt.xlabel('Epoch')
plt.ylabel('Validation Accuracy')
plt.legend()
plt.title('ANN vs CNN Validation Accuracy')
plt.show()
```

```
ANN Test Accuracy: 0.4756
CNN Test Accuracy: 0.7140
Improvement with CNN: 0.2384
```



Discussion: CNN outperforms ANN because it uses convolutional layers to capture spatial features (e.g., edges, textures) in images, while ANN flattens the input and loses this information, leading to lower accuracy (typically ANN ~45-50%, CNN ~70-75% on CIFAR-10 with this setup).